Most of the world's servers run on Linux based operating systems and the logs are usually stored in the */var/log* folder.

For example, if you want to view who requested *sudo* privileges to your system, you can view the *auth.log* file in this folder.

All the logs placed side by side generate a story that tells who and which applications did exactly what on the computer without any ambiguity.

The logs of a Linux machine act in exactly the same way as a log register you would use for noting down every small step you take. It's similar to a photo album that records how a child grows up gradually.

Logs generated by intrusion detection and prevention systems (IDPS) are specifically important for maintaining the security of systems. For example, Figure 5 shows an example of a server protected by the IDPS Suricata, which is under a possible SSH brute force attack.

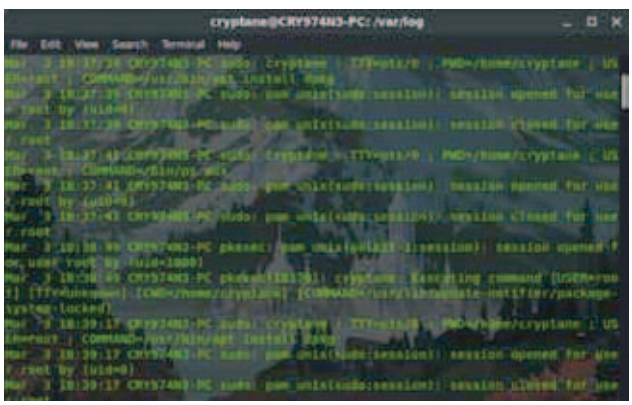

Figure 4: Logs stored in *auth.log*

The IDPS logs are important to understand how an attacker might be exploiting the flaws in the network to gain access to hosts. The vulnerabilities need to be patched as and when they are spotted.

There are various automated tools for visualising the logs, one of them being the ELK (Elasticsearch, Logstash, Kibana) stack. It can form easy-to-read dashboards to visualise, analyse and find anomalies in the logs.

Some attackers may try to modify and erase the logs, but those operations also get stored. It is possible to make the story difficult to read but it is never possible to erase it.

Logs, in the end, are very useful when it comes to telling the story of the machine, especially from the perspective of cyber security. To a lay person using the computer, logs may seem irrelevant. We do not see many people opening the Event Viewer in Windows or the log folder in Linux. But then again, when it comes to an incident response, the first step is always to analyse the logs. Because the logs always have a story to tell to those who can read it. END

By: Anisha Ghosh and Aditya Mitra

The authors are cyber security researchers with an interest in networking and Nix based systems. They usually spend their leisure time managing servers and deployments.

Figure 5: Suricata logs